

Cache-Aware Request Routing for Distributed RAG-Based LLM Serving: Design, Calibration, and Empirical Findings

Saadet Cansu Baktiroğlu
Dept. of Artificial Intelligence
Engineering
TOBB University
of Economics and Technology
Student ID: 231401023
sbaktiroglu@etu.edu.tr

Umut Baran Boztaş
Dept. of Computer Engineering
TOBB University
of Economics and Technology
Student ID: 231101064
uboztas@etu.edu.tr

Demir Doruk Dilek
Dept. of Computer Engineering
TOBB University
of Economics and Technology
Student ID: 231104088
demirdoruk.dilek@etu.edu.tr

Abstract—Retrieval-augmented generation (RAG) over replicated large language model (LLM) servers creates a joint decision: the router must select a replica and order the retrieved chunks so that their prefix matches that replica’s KV cache. We implement this decision in an OpenAI-compatible router for two vLLM workers and evaluate five core strategies under calibrated cache capacity, cold starts, rotated arm order, 800-request traces, and three repetitions. Cache-blind policies reuse 55.5–56.2% of prompt tokens, flat cache-aware policies reach 66.1–66.4%, and the worker-specific joint strategy `per_worker_tree` reaches $76.1 \pm 0.06\%$, a 9.7 percentage point gain over `cache_aware`. The gain grows to 12.0 points at the highest replicated load and falls to 4.2 points on a low-locality trace, consistent with the proposed cache-locality mechanism. Its reported $10.7\times$ median-TTFT advantage at that load is only a completion-path measurement, however, and is not an end-to-end speedup. Repeated controls do not separate `cache_aware` from the local `DualMap`-inspired prototype, nor fixed from overlap-adaptive routing; a semantic-labelled arm is a protocol no-op at two workers. Recalibrating tracker capacity preserves the earlier ordering ranking but eliminates the previously reported beta-by-load/locality crossover. Dispatch-time cache bookkeeping remains more rank-accurate under load than completion-time bookkeeping. We report these negative replications, prompt-serialization and raw-artifact limitations explicitly, and add sweep tooling that records end-to-end timing, p99 latency, load coefficient of variation, and file-backed router logs.

Index Terms—LLM serving, KV cache, prefix caching, request routing, retrieval-augmented generation, load balancing, vLLM

I. INTRODUCTION

Serving large language models behind a pool of replicas is now standard practice for throughput and availability. When the workload is retrieval-augmented – each request’s prompt is built from a fixed system preamble followed by a handful of retrieved document chunks – the serving engine’s prefix KV cache can eliminate most of the prefill cost for a request *if* it is routed to a replica that already holds a matching prefix in cache. A router that is blind to this fact (round robin, or load balancing on queue depth alone) discards a large, freely available source of latency reduction: which replica currently

holds which prefix is entirely a function of routing history, so the router itself controls the very signal it would need to route well.

This paper describes a cache-aware router built as a transparent reverse proxy in front of N vLLM [2] replicas. Its starting point is a shared scoring function,

$$\text{score}(w) = \alpha \cdot \text{cache_gain}(w) - \beta \cdot \text{load}(w),$$

of which round-robin, pure load balancing, and cache-aware routing are special cases. The central extension keeps a separate history tree for each worker and evaluates that worker’s best cache-friendly chunk order jointly with its load, rather than computing one global order and then routing it.

The evaluation makes four contributions:

- 1) **Calibrated system comparison.** A five-arm, three-run sweep compares two cache-blind baselines, two flat cache-aware policies, and worker-specific joint routing/reordering. At speedup eight, `per_worker_tree` raises aggregate cached-token fraction from $66.4 \pm 0.4\%$ to $76.1 \pm 0.06\%$ relative to `cache_aware`, while contains-gold quality remains within the 77.5–77.8% band.
- 2) **Mechanism-oriented axes.** Repeated load and locality sweeps show a 9.7–12.0 point gain at the two replicated load points, but only 4.2 points under low locality. This conditionality is more informative than a single winner at one operating point.
- 3) **Negative replications.** Three runs do not separate `cache_aware` from `cacheweaver_dualmap`, fixed from overlap-adaptive routing, or the reported semantic-labelled arm from its non-semantic parent. Calibrated beta sweeps do not reproduce the earlier crossover pattern. We treat “not separated” as an uncertainty statement, not equivalence.
- 4) **Measurement controls.** The cache tracker is calibrated to 5,840 blocks; arm order is rotated; engines and

router are cold reset; schedule lag, p99 latency, load CV, quality, decision time, and end-to-end timing are recorded. File-backed Uvicorn logs remove an unread-pipe deadlock that had been misdiagnosed as strategy overload.

The result also requires two qualifications. First, the historical `per_worker_tree` timings start after the preliminary decision call and therefore cannot establish an end-to-end latency advantage. Second, that two-phase run used a chunk separator adding 45 prompt tokens per request, producing a 1.9% denominator mismatch relative to one-phase arms. The revised harness fixes the serialization and records both completion-path and end-to-end timing, but the reported measurements predate that fix. Our claim is therefore about the measured cache-fraction separation under the stated protocol, not a protocol-neutral latency speedup. The central question is correspondingly conditional: when does worker-aware ordering add value, and how does that value change as load and retrievable overlap change?

II. RELATED WORK

Three largely separate literature threads intersect with this work: distributed cache-affinity routing, RAG-specific cache reuse, and adaptive/semantic parameter tuning. Table I positions the systems most relevant to the implementation; vLLM [2] underlies the serving engine and is discussed separately at the end of this section.

A. Distributed Cache-Affinity Routing

Preble [6] routes requests sharing a prefix to the same replica to maximise KV-cache reuse, maintaining a precise prefix tree per replica and switching from prompt-aware to load-aware scheduling once a request’s prefix-hit rate falls below a threshold. Mooncake [7] disaggregates prefill and decode into separate clusters around a KVCache-centric scheduler that balances throughput against latency SLOs. NVIDIA Dynamo [8] implements a production KV-cache-aware router that scores replicas on a combined decode-cost/prefill-cost estimate derived from cache overlap. DualMap [4] identifies a shared weakness across this line of work – oscillating between a single cache-affinity objective and a single load-balancing objective in one shared mapping space – and proposes a “power of two choices”-style fix: each request is mapped to two independent candidate replicas via two independent hashes, with SLO-aware routing and hotspot-aware rebalancing arbitrating between them.

This thread solves worker selection but is *not RAG-specific*: it treats the prefix as a fixed session history and does not address the fact that retrieval returns a *set* of documents whose order can differ from one query to the next even when the set itself overlaps heavily with a previous query’s.

B. RAG-Specific Cache Reuse

RAGCache [9] builds a knowledge tree over retrieved document sequences and caches intermediate states across the GPU/host memory hierarchy, additionally overlapping

retrieval with speculative generation. CacheBlend [10] and TurboRAG [11] target the case where reusable chunks do *not* form a contiguous prefix at all, fusing precomputed KV caches (selectively recomputing only a small subset of tokens to preserve generation quality) rather than relying on prefix alignment. Cache-Craft [12] shows that plain prefix caching degrades substantially on production RAG traffic and proposes identifying which chunk-caches remain valid, cheaply repairing the ones that do not. CacheWeaver [3] – the mechanism this paper’s `per_worker_tree` strategy builds on – takes a different, engine-unmodified approach: a lightweight prompt-layer method that reorders retrieved chunks into a cache-friendly sequence via a greedy prefix-tree walk (its Algorithm 1), reported to reach 97.5% of an oracle ordering’s reusable-prefix depth.

This thread addresses RAG-specific cache reuse; CacheWeaver is the member that specifically reorders retrieved evidence. The listed systems assume a single serving instance and do not include distributed worker selection.

C. Adaptive Tuning and Semantic Approaches

GORG0 [13] models per-request cost as the sum of network, queueing, and prefill terms and tunes routing weights online via an evolutionary strategy against a held-out tuning window; it is network-topology-aware but not RAG-specific and requires black-box search rather than a closed-form update. The semantic-caching family – GPTCache [14], SCALM [15], and MeanCache [16] – compares query embeddings and, on a sufficiently similar match, returns a cached *answer* directly, skipping computation entirely; this operates one layer above KV-cache reuse or worker routing and carries a distinct risk profile (a false-positive match returns a wrong answer outright, rather than merely costing extra prefill).

D. Positioning of This Work

TABLE I
RELATED WORK COMPARED ALONG THREE AXES

System	Dist. routing	RAG cache reuse	Adaptive
Preble [6]	✓	×	partial (threshold)
Mooncake [7]	✓	×	×
Dynamo [8]	✓	×	×
DualMap [4]	✓	×	×
RAGCache [9]	×	✓	×
CacheBlend [10]	×	✓	×
TurboRAG [11]	×	✓	×
Cache-Craft [12]	×	✓	×
CacheWeaver [3]	×	✓	×
GORG0 [13]	✓	×	✓ (evolutionary)
Semantic caching	×	×	×
This work	✓	✓	✓ (overlap-adaptive)

This paper combines distributed routing (Section III-A/B/D) with RAG-specific reordering (Section III-C) *in one system*: retrieval output is first reordered for cache friendliness, and that ordering is then evaluated per-replica and optimised jointly with worker selection, rather than as two independent stages. To our knowledge no system in the surveyed literature

addresses this specific dependency – whether to reorder before or after choosing a worker changes which ordering is actually optimal, since two replicas’ caches generally disagree on what is cache-friendly (Section III-C). Techniques from Section II-C above (adaptive tuning and semantic selection) are borrowed as extended evaluation components layered on top of this core design, not presented as this paper’s primary contribution.

RAGRoute [5] is related in spirit but orthogonal in target: it uses a lightweight classifier before retrieval to choose which federated data sources to query. It does not decide which LLM replica’s KV cache is most relevant to an incoming query. Section III-F describes the prototype semantic pre-filter explored for that distinct decision.

Serving infrastructure. vLLM [2] provides the underlying PagedAttention KV-cache engine and the OpenAI-compatible API surface this router proxies. All prefix-cache accounting in this work is calibrated against vLLM’s own reported counters (`prompt_tokens_details.cached_tokens` and the Prometheus `vllm:prefix_cache_*` series) rather than assumed.

III. SYSTEM DESIGN

A. Architecture

The router is a FastAPI reverse proxy exposing the same `/v1/chat/completions`, `/v1/completions`, and `/v1/models` surface as vLLM itself, so that a client (in our deployment, an Open WebUI instance) cannot distinguish it from talking to a bare engine – streaming responses are forwarded byte-for-byte rather than buffered. A background poller scrapes each worker’s `/metrics` endpoint on a fixed interval and maintains a per-worker `WorkerState` snapshot (queue depth, running-request count, KV-cache utilisation, prefix-cache hit counters); routing decisions read this snapshot rather than querying workers synchronously on the request path.

Because vLLM exposes aggregate cache statistics but not “do you currently hold a cache entry for this specific prefix”, the router keeps its own approximation: for each worker, an LRU of block-aligned prefix hashes it has been sent (`PrefixTracker`), hashed in the same fixed token-block granularity vLLM itself uses. Two independent bookkeeping conventions are supported for *when* a block is recorded as cached – at dispatch time (optimistic: the block will be resident once prefill starts) or at completion time (conservative: only once the engine has actually finished) – and Section VI-F reports the result of ablating between them.

B. Strategy Framework

All registered strategies share a common interface, `select(context, snapshot) -> Decision`, so that switching the active policy is a configuration change rather than a code change. Five policies use the ordinary one-phase path:

- **round_robin:** $\alpha=\beta=0$, plus a rotating tie-break; the floor every other strategy is measured against.

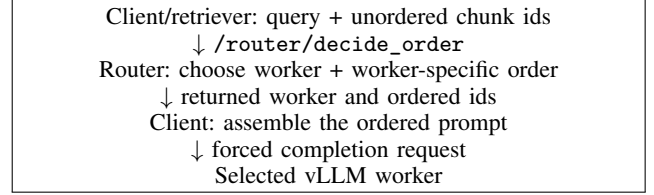


Fig. 1. Current client-mediated two-phase path. The client applies the returned order. The revised replay harness records decision, completion-path, and end-to-end timings separately; the historical results in Section VI used only the completion-path timer.

- **least_loaded:** $\alpha=0$; pure, cache-blind load balancing on a saturating load-normalisation $\text{load_norm}(w) = \text{load}(w)/(\text{load}(w) + \text{LOAD_REF})$, chosen over a clipped $\min(\cdot, 1)$ form because clipping destroys the policy’s discriminative power once both replicas exceed the reference load (Section IV-C).
- **cache_aware:** the full scoring function plus an *adaptive guard band*, $\delta = \delta_0 \cdot (1 - \text{kv_usage})$; the winning replica by score is overridden back to the least-loaded replica if its load exceeds the minimum by more than δ . This prevents the classic failure mode of pure cache-affinity routing, where the replica holding a popular prefix keeps attracting traffic until its queue makes it the worst choice by latency despite remaining the best choice by cache.
- **adaptive_cache_aware:** `cache_aware` with β and δ_0 driven by an overlap EWMA, while a CUSUM detector records diagnostic drift alarms (Section III-E).
- **cacheweaver_dualmap:** the DualMap-inspired two-choice prototype described in Section III-D.

The two remaining registry entries, `per_worker_tree` and `semantic_per_worker_tree`, expose the separate decision capability described next. This registry taxonomy must not be confused with the arms of any one experiment: Section VI-A compares complete system strategies, while Section VI-D isolates order under fixed routing.

C. Worker-Specific Joint Reordering (`per_worker_tree`)

DualMap’s replica candidate-selection step (its Eq. 5) hashes the reordered chunk sequence through two independent consistent-hashing rings to obtain a candidate pair (r_1, r_2) , with a collision rule $r_2 \leftarrow (r_1 + 1) \bmod n$ when the two rings agree. At $n=2$ replicas – the deployment size used throughout this work – this collision rule is not an edge case: it is provably always triggered or already trivially satisfied, so $\{r_1, r_2\} = \{0, 1\}$ on every request. The dual-hash candidate-selection machinery is therefore a mathematically guaranteed no-op at this scale, and any cache-affinity signal has to come from elsewhere in the pipeline.

`per_worker_tree` replaces the hash-and-select step entirely. Each worker maintains its own independent CacheWeaver knowledge tree; for every candidate worker, `best_reorder_for()` computes both that worker’s own best reordering of the retrieved chunks *and* a normalised cache-gain figure against that specific tree, and the scoring function from Section III-B picks the winner directly – there is no

candidate-pruning step to degenerate. This is a genuine change in what "the right chunk order" means: it is no longer a single, worker-independent quantity computed once and then assigned to whichever replica is chosen, but something that emerges jointly with the worker decision, since the two replicas' trees generally disagree about which order is cache-friendly.

Because the winner and its optimal chunk order are decided together, a client-mediated two-step protocol is used (Fig. 1). The client first calls `/router/decide_order` with the unordered, retrieval-order chunk ids and receives back `{worker, ordered_chunk_ids}`; it then builds the prompt in that returned order and issues the completion with an `x-router-force-worker` header. The router forwards this already ordered prompt to the named worker. Phase-one bookkeeping is optimistic: the tree is updated before phase two succeeds, so the two HTTP calls are not transactional.

1) *Token-weighted cache gain*: The router module has an optional token-weighted path, $\text{gain} = \text{hit_tokens}/\text{total_tokens}$, for cases where the caller supplies a chunk-id-to-text mapping. It reuses the existing tokenizer and a synthetic regression test confirms that few large chunks and many small chunks can select different workers. This path is *not live-integrated*: `/router/decide_order` currently accepts only chunk ids, and the strategy calls the module without chunk texts. Consequently each chunk has weight one on the live HTTP path and all reported live `per_worker_tree` traffic remains chunk-count-based. Section VI-G is therefore an isolated functional test, not a measured live improvement.

2) *Quality protection (`protect_top_k`)*: Pure greedy reordering can push a highly relevant but uncached chunk to the end of the prompt while pulling a barely relevant but cached chunk to the front – a risk CacheWeaver's own evaluation does not fully rule out (its Table 7 states only that reordering *did not measurably hurt* answer quality in their setting, not that it never can). `protect_top_k` pins the first K retrieval-order chunks in the returned list and reorders only what follows; $K=0$ reproduces the original behaviour. A regression test verifies this positional property. It does not establish that the post-pinned traversal remains one cache-contiguous prefix when a pinned chunk is absent from the tree, nor does it establish answer quality; both require end-to-end validation.

D. The `cacheweaver_dualmap` Baseline and a Tie-Break Defect

DualMap's official implementation (<https://github.com/ASISys/DualMap>, MIT-licensed, which also incorporates Preble) is publicly available, but was not used directly: it is written against an 8+-GPU, 512 GB-DRAM reference deployment and consumes Mooncake-format traces (fixed prefixes, no real retrieval step), neither of which matches this project's two-machine RAG deployment with genuine per-query retrieval. The local `cacheweaver_dualmap` strategy is therefore a *DualMap-inspired two-choice prototype*, not an equivalent reimplement. It uses two independent BLAKE2b-modulo mappings rather than consistent-hash rings, estimates recompute cost with a fixed 200-token-per-chunk placeholder,

and derives pending-token load from vLLM's waiting-request gauge. On the documented vLLM V1 setup that gauge can remain zero under load, so SLO/rebalancing activation has not been demonstrated. Moreover, although the inner prototype computes a CacheWeaver order, the enclosing strategy does not propagate that order to the proxy's prompt-rewrite field; vLLM therefore receives the original prompt order in the reported one-phase runs. Results bearing this strategy name evaluate its current two-choice routing and internal bookkeeping, not live CacheWeaver reordering or a faithful DualMap run.

An earlier internal version also used one shared tree for both replicas, making their recompute estimates identical. Per-replica trees fixed that error but exposed a separate tie defect: non-strict comparisons always selected the hash-assigned r_1 when estimates tied. The current implementation randomises this candidate tie; Section VI-G describes the corresponding GPU-free regression test.

E. Overlap-Adaptive Weighting

A common formula underlies the adaptive modes: Jaccard overlap $J(A, B) = |A \cap B|/|A \cup B|$. The current call sites do *not*, however, observe the same stream. `adaptive_cache_aware` compares adjacent requests within each session, whereas `per_worker_tree`'s modes compare globally adjacent requests and initialise the first comparison against an empty set. The offline tool reports both notions separately (Section IV-D); sharing a formula does not make their calibrated values interchangeable.

Threshold mode. A one-line switch: $\alpha=0.2$, $\beta=0.8$ when J falls below a configured threshold (locality is a weak signal, weight load instead), and $\alpha=0.8$, $\beta=0.2$ otherwise, with $\alpha + \beta=1$ by construction.

EWMA with CUSUM diagnostics. An exponentially weighted moving average tracks a smoothed overlap estimate $D_t = \lambda J_t + (1 - \lambda)D_{t-1}$. This EWMA, together with the fixed configured D_{target} , feeds $\beta_{\text{live}} = \beta_{\text{base}} \cdot \text{clip}\left(1 + k \cdot \frac{D_{\text{target}} - D_t}{D_{\text{target}}}, [0.3, 3.0]\right)$, while α is held fixed. A CUSUM detector is updated alongside it and recalibrates its own alarm reference, but the returned alarm and recalibrated reference are not consumed by routing. CUSUM is therefore observational telemetry in the current code, not a second decision input. Offline calibration now reconstructs the exact arrival-ordered, within-session stream observed by `adaptive_cache_aware`; at retrieval depth ten this gives $D_{\text{TARGET}}=0.322$ and favours `CUSUM_H=0.30` over the previous 0.20 default. The per-worker mode still consumes global adjacency, so the calibrated target is not transferable to that path. GPU-free tests verify that off reproduces the fixed baseline and that threshold/EWMA inputs can change a synthetic decision. The live hot and drifted comparisons in Section VI-C find no outcome separation at three repetitions.

F. Semantic Candidate Pre-Filtering

For deployments with more than a handful of replicas, `per_worker_tree`'s $O(k^2)$ per-candidate reorder becomes

$n \cdot O(k^2)$ across all healthy workers – cheap at $n=2$, not necessarily cheap beyond it. `SemanticWorkerRouter` addresses this independently of locality-based weighting: each worker’s recent query traffic is summarised as an EWMA centroid in embedding space (`multilingual-e5-base`, the same checkpoint used for corpus retrieval, loaded once per router process); an incoming query is embedded and ranked by cosine similarity against each worker’s centroid, and only the top- k ranked workers are passed to `per_worker_tree`’s comparison. A worker with no observed traffic yet is assigned the *average* similarity of workers that do have a centroid – neither a fixed low score (which would permanently exclude new or recently restarted workers) nor a fixed high score (which would make every cold worker an automatic favourite) – and the candidate set falls back to every healthy worker whenever the semantic favourites happen to be unhealthy, so the mechanism cannot starve a request. Loading the embedding model is forced synchronously at router start-up rather than being paid by whichever live request happens to arrive first.

The reported semantic-labelled arm cannot exercise the intended pre-filter: `SEMANTIC_TOP_K=2` equals the entire two-worker pool, so no candidate is removed. The revised two-phase contract now carries query text and the experiment runner can request top- $k=1$; those changes make a future active ablation possible but do not retroactively turn the reported no-op into a semantic result. A scalable evaluation still requires $N > k$ and must include embedding overhead in end-to-end timing.

IV. IMPLEMENTATION DETAILS

A. Corpus and Trace Construction

The RAG corpus is built from the publicly available TQuAD dataset [1], a SQuAD-format Turkish question-answering resource (681 articles, 8,308 questions). Each question already points at exactly one source paragraph, so using the paragraph as the retrieval chunk gives retrieval ground truth "for free". Paragraphs are split at sentence boundaries only when they exceed a maximum chunk length, and every chunk is padded (in token space, not by appending raw text – tokenizers merge whitespace unpredictably) to a multiple of vLLM’s KV-cache block size, since a chunk boundary that does not align with a block boundary produces a different block hash even when the underlying content is shared, silently breaking the prefix chain the whole system depends on. The resulting corpus contains 2,619 chunks. Before padding, lengths range from 9 to 754 tokens (mean 182.2, $\sigma=108.5$); after padding they range from 16 to 768 tokens (mean 189.6, $\sigma=108.4$). The variation motivates the token-weighted prototype of Section III-C.1, but that prototype is not yet connected to live requests.

Synthetic request traces are generated once per experiment and replayed identically across every strategy under comparison, so that strategies are never compared against different workloads. Two independent knobs control cache locality: popularity skew across titles (a Zipf distribution with configurable exponent s) decides *which* chunks are asked about

repeatedly, and session structure (several questions drawn from one title, arriving with realistic think-time gaps) decides *when* repeats occur relative to each other – a popular chunk asked about ten times over an hour is not locality if the cache empties between hits. An optional popularity-drift knob rotates the ranking over the course of a trace, exercising the scenario an adaptive policy is meant to handle and a static one is meant to degrade gracefully under.

B. Software and Deployment Platform

The implementation is written in Python and uses FastAPI and Uvicorn for the OpenAI-compatible reverse proxy, `httpx` for asynchronous stream forwarding and worker polling, and `prometheus-client` for router instrumentation. Each upstream runs vLLM with Qwen2.5-7B-Instruct; the two workers communicate with the router over a private Tailscale network. Hugging Face Transformers supplies the serving-model tokenizer, `multilingual-e5-base` supplies corpus and query embeddings, and NumPy stores and scores the offline embedding matrix. The submission also contains replay, trace-generation, calibration, quality-scoring, schedule-lag, and GPU-free regression scripts. Runtime parameters, worker addresses, strategy selection, and calibration constants are exposed as environment variables rather than requiring source edits.

C. Calibration Discipline

TABLE II
CALIBRATED SETTINGS AND DOCUMENTED EXPERIMENTAL EXCEPTIONS

Constant	Value	Measured via
LOAD_REF	16 concurrent requests	Kleinrock power (throughput/latency) peak, single-worker concurrency sweep, Qwen2.5-7B, ~ 1700 -token prompts
Mean prompt size	2,392 tokens	Mean prompt token count, $n=800$ live RAG requests, top- $k=10$
DualMap-inspired SLO / rebalance thresholds	38,272 / 57,408 tokens	LOAD_REF $\times$ mean prompt tokens; $1.5\times$ ratio preserved from DualMap’s relationship
D_TARGET (drift reference)	0.322	Arrival-ordered within-session Jaccard stream, top- $k=10$, 2,170 stable observations
CUSUM_H	0.30	Grid calibration on nominal and drift-0.1 traces; 26.73 nominal alarms/1,000 and $2.20\times$ separation
TRACKER_CAPACITY	5,840 blocks	Smaller worker’s KV-cache capacity: $\min(93,440, 93,840)/16$

Table II records the values used by the main strategy, load/locality, calibrated-ordering, and calibrated-beta experiments. Semantic top- k , centroid learning rate, CUSUM margin K , EWMA rate, and `protect_top_k` retain explicit prototype defaults. The source default for `TRACKER_CAPACITY` is 50,000; calibrated routing experiments require `ROUTER_TRACKER_CAPACITY=5840`.

TABLE III
EXPERIMENT FAMILIES AND PROTOCOL SCOPE

Family	Protocol
Core strategies	800 requests, top- $k=10$, speedup 8, five arms, three repetitions, capacity 5,840
Load/locality	cache_aware vs. per_worker_tree; speedups 4/8/12 and high/low synthetic locality
Adaptive/semantic	Three repetitions on hot and drift-0.1 traces; semantic arm retains all two candidates
Ordering	Four fixed-router orders, top- $k=10$, three repetitions; original 50,000 and calibrated 5,840 capacities
Beta	High/low locality $\times$ speedups 30/35 $\times \beta \in \{0, 1\}$; one retained result per cell at capacity 5,840
Bookkeeping	Dispatch vs. completion, speedups 5/15, three repetitions; historical capacity 50,000

Only the historical dispatch/completion sweep and the original ordering run used 50,000; both held it fixed across arms and are labelled as historical exceptions. Section VI-D reports the ordering replication at 5,840.

D. Overlap Characterisation Tool

A standalone tool computes the Jaccard overlap between consecutive requests’ retrieved chunk sets directly from a corpus and trace, without touching the router or a live engine (retrieval alone is enough). It reports two distinct notions of "consecutive" separately, because they answer different questions: *session-adjacent* overlap (turn k to turn $k+1$ within one session) reflects what a prefix cache actually sees if the router keeps a session pinned to one worker, while *global-adjacent* overlap (arrival-order adjacency, ignoring session boundaries) reflects the raw opportunity the router faces before any session-affinity policy is applied. This distinction turns out to matter a great deal (Section VI-G).

V. EXPERIMENTAL SETUP

All live experiments use two RTX 4090 workers running Qwen2.5-7B-Instruct with GPU-memory utilisation 0.85; one is local and the other is reached over Tailscale. Requests are released at their trace offsets without waiting for earlier completions. Before each arm, both engines and the router are restarted to clear their cache and tracker state. Arms are interleaved repeat-major and rotated across repetitions. The router’s estimated cached tokens are checked against vLLM’s `prompt_tokens_details.cached_tokens` response field.

The high-locality trace uses Zipf $s=1.5$ and session length four; the low-locality trace uses $s=0.7$ and session length two. The main 15 runs have schedule-lag p99 at most 2 ms. At the selected operating point, tracker validation gives Pearson correlation 1.000, mean absolute error 0.6 token, and zero underestimation. Speedup eight avoids both the weak top- $k=3$, speedup-five regime, where strategies cluster, and speedup 20, where median TTFT reaches 8.8 s and queueing dominates.

For each run, cache reuse is $\sum \text{cached prompt tokens}$ divided by $\sum \text{prompt tokens}$. We report TTFT p50 and p99,

request-count load CV including zero-count workers, and normalised output “contains gold” as the primary quality check. Means use the sample standard deviation over repetitions. As a descriptive separation rule, a mean difference must exceed twice the larger arm standard deviation; failing that rule means unresolved, not equivalent. The historical ordering scorer uses the mean of per-request cache fractions instead of the run-level aggregate and is labelled separately.

The reported two-phase `per_worker_tree` runs time only the completion request. Their decision call, prompt construction, and first HTTP round trip are excluded. They also use a separator five tokens longer at each of nine boundaries, adding about 36,000 prompt tokens over 800 requests (35,997 observed, 1.9%). The revised replay code uses matched serialization and records decision, completion, and end-to-end fields, but the historical numerical results were not rerun after that fix.

VI. RESULTS

Artifact status. The repository includes the overlap scores, trace manifests, experiment drivers, calibration tools, regression tests, and a configuration-recording sweep harness. It does not retain the raw live-run JSONL outputs underlying the numerical live tables. Those values are preserved in the project log and final report but cannot be recomputed from the submitted tree without rerunning the two vLLM workers.

A. Calibrated Core Strategy Comparison

Table IV supplies the paper’s previously missing system-level comparison. Flat cache-aware routing adds 10.9 points over `least_loaded`; worker-specific joint routing and ordering adds another 9.7 points over `cache_aware`. Both clear the descriptive spread threshold (1.6 and 0.9 points, respectively). The five quality means occupy a narrow 77.5–77.8% band. Conversely, `cache_aware` and `cacheweaver_dualmap` do not separate on cache reuse, p50, or p99; the earlier single-run latency difference was noise at this replication level. The `per_worker_tree` p99 gap also narrowly fails the spread rule, and its p50/p99 values must not be read as end-to-end comparisons because of the two-phase timer.

B. Load and Locality Characterisation

Figure 2 shows that the cache advantage remains 9.7–12.0 points at the replicated $8\times$ and $12\times$ loads. At $12\times$, median completion-path TTFT is 0.226 s versus 2.411 s for the one-phase baseline, a $10.7\times$ ratio that is consistent with less prefill work preventing queue growth but is not an end-to-end speedup or a causal decomposition. The $4\times$ direction is similar (+10.2 points) but has only one run.

Locality provides the sharper mechanism check. On the high-locality trace at speedup 12, the cache fractions are $76.6\pm0.3\%$ and $64.6\pm0.5\%$; on the low-locality trace they fall to $57.3\pm0.3\%$ and $53.1\pm0.1\%$. The gain therefore contracts from 12.0 to 4.2 points, as a cache-reuse mechanism predicts. Low-locality p99 is unresolved. Its 1.2-point contains-gold difference is directionally consistent across runs but is not

TABLE IV
CALIBRATED FIVE-STRATEGY COMPARISON AT SPEEDUP 8 (MEAN $\pm$ SAMPLE STANDARD DEVIATION, THREE RUNS; CACHE IS AGGREGATE CACHED/PROMPT TOKENS).

Strategy	Cached fraction	TTFT p50	TTFT p99	Load CV	Contains gold
least_loaded	55.5 $\pm$ 0.8%	0.785 $\pm$ 0.126 s	3.368 $\pm$ 0.739 s	0.008	77.8 $\pm$ 0.3%
round_robin	56.2 $\pm$ 0.2%	0.742 $\pm$ 0.063 s	2.564 $\pm$ 0.082 s	0.000	77.7 $\pm$ 0.3%
cacheweaver_dualmap	66.1 $\pm$ 0.6%	0.484 $\pm$ 0.303 s	2.633 $\pm$ 0.679 s	0.358	77.7 $\pm$ 0.1%
cache_aware	66.4 $\pm$ 0.4%	0.278 $\pm$ 0.028 s	2.320 $\pm$ 0.645 s	0.260	77.5 $\pm$ 0.1%
per_worker_tree	76.1$\pm$0.06%	0.147 $\pm$ 0.001 s [†]	1.184 $\pm$ 0.190 s [†]	0.196	77.6 $\pm$ 0.15%

[†] Completion-path timing; the decision call and first HTTP round trip are excluded.

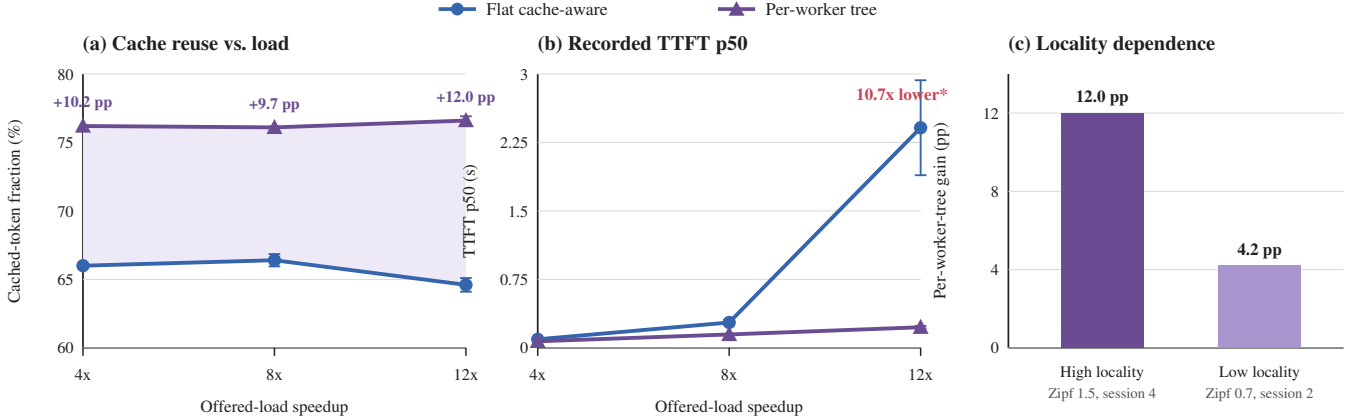


Fig. 2. Worker-specific routing across load and locality. At replicated speedups 8 and 12, **per_worker_tree** improves cached-token fraction over **cache_aware** by 9.7 and 12.0 percentage points; the 4 $\times$ point has $n=1$ and is exploratory. Reported **per_worker_tree** TTFT is completion-path timing, not end to end. Its historical prompt separator also adds 35,997 prompt tokens (1.9%) over 800 requests, so cache denominators are not perfectly matched.

TABLE V
REPLICATED CONTROLS (CACHED-TOKEN FRACTION)

Comparison	Arm A	Arm B
Adaptive / fixed, hot	66.5 $\pm$ 0.5%	66.4 $\pm$ 0.5%
Adaptive / fixed, drift 0.1	43.8 $\pm$ 0.7%	44.0 $\pm$ 0.5%
Semantic-labelled / PWT, hot	76.1 $\pm$ 0.40%	76.1 $\pm$ 0.06%

TABLE VI
WORKER-BLIND ORDERING AT ORIGINAL AND CALIBRATED CAPACITIES

Order	50,000 blocks	5,840 blocks	Change
Greedy	73.1 $\pm$ 0.2%	73.2 $\pm$ 0.3%	+0.1 pp
Relevance	67.7 $\pm$ 0.1%	67.6 $\pm$ 0.4%	-0.1 pp
Canonical	65.0 $\pm$ 0.5%	64.9 $\pm$ 0.2%	-0.1 pp
pinned_prefix	62.9 $\pm$ 0.2%	62.7 $\pm$ 0.4%	-0.2 pp

claimed as a quality improvement without a paired question-level test.

C. Repeated Null Results

The adaptive policy’s β is active (405 distinct values spanning 0.65–1.28), yet neither the hot nor drifted trace separates from fixed **cache_aware**. CUSUM cannot explain this null result because its alarms are telemetry and are not consumed by the router. The reported semantic arm is a structural no-op: top- $k=2$ retains both workers, so its identical hit rate says nothing about semantic usefulness. The revised query-carrying protocol and top- $k=1$ option require a fresh measurement.

D. Ordering Result Survives Capacity Calibration

This ablation holds **cache_aware** routing fixed and changes only one global, worker-blind prompt order. Calibration moves every mean by at most 0.2 points and preserves the ranking:

greedy is 5.6 points above relevance and 8.3 above canonical at 5,840 blocks. Thus a rule that observes estimated cache state wins; canonical and session-prefix ordering do not. At top- $k=3$, a single preliminary run spans only 2.4 points, so the lever opens with retrieval depth. The result uses the historical request-macro metric and must not be numerically conflated with Table IV’s run-level aggregate.

E. Calibrated Beta Matrix

Here $\beta=1$ includes the load term and $\beta=0$ is pure cache routing. The earlier uncalibrated crossover pattern does not replicate: $\beta=1$ is lower in three cells and the remaining difference is 4 ms. Because per-cell spreads are absent from the supplied result bundle, Table VII reports directions rather than three significant wins. Cache fraction is nearly unchanged across beta; the observed differences lie in queueing behaviour.

TABLE VII
TTFT P50 IN THE CALIBRATED BETA MATRIX

Locality	Load	$\beta = 1$	$\beta = 0$	Direction
High	30×	0.175 s	0.484 s	$\beta=1$
High	35×	1.323 s	2.734 s	$\beta=1$
Low	30×	0.151 s	0.147 s	unresolved
Low	35×	0.442 s	0.940 s	$\beta=1$

TABLE VIII
CACHE-STATE ACCURACY AT TWO REPLAY LOADS (THREE RUNS PER MODE)

Spd.	Mode	Corr.	Rank agree.	Underest.
5×	Dispatch	0.972±0.010	97.1±0.6%	0.0±0.0%
5×	Completion	0.941±0.011	92.5±0.6%	6.2±0.7%
15×	Dispatch	0.980±0.003	98.0±0.0%	0.0±0.0%
15×	Completion	0.806±0.008	79.7±0.9%	29.6±0.8%

F. Bookkeeping and Drift Diagnostics

Completion-time bookkeeping misses related requests that are still in flight; as load rises, its correlation falls from 0.941 to 0.806 and underestimation rises from 6.2% to 29.6%, while dispatch remains near 0.98 correlation. Since this historical sweep uses 50,000 blocks, it supports the relative load mechanism, not calibrated absolute error.

Trace calibration changes D_TARGET from 0.529 to 0.322 and CUSUM_H from 0.20 to 0.30. The latter reduces nominal alarms from 108.76 to 26.73 per 1,000 and improves drift/nominal separation from 1.52× to 2.20×. A nominal trace has no labelled change points, however, and CUSUM alarms do not affect routing; this remains diagnostic telemetry.

G. Harness Corrections and Isolated Checks

The sweep launcher previously left Uvicorn stdout/stderr connected to an unread pipe. Its 64-KB buffer eventually filled and blocked the router, so the old “two-phase overload” diagnosis is withdrawn. Each run now writes router output to its own log file. The shared scorer now uses aggregate prompt-token weighting, inclusive p99, and load CV over all expected workers; the replay records decision and end-to-end fields and uses one separator across one- and two-phase paths.

A randomized tie replaces the DualMap-inspired arm’s deterministic first-candidate choice; a 400-request GPU-free test checks that neither worker falls below 35%. A separate synthetic test shows why optional token weighting matters: two cached chunks totalling about 1,300 tokens should beat three cached chunks totalling about nine tokens, although the live decision endpoint still lacks chunk texts. Finally, offline overlap remains strongly session-structured: on the baseline 3,000-request trace, mean session-adjacent Jaccard is 0.476 versus 0.055 for global adjacency. These checks explain design choices but are not additional live performance claims.

VII. DISCUSSION AND LIMITATIONS

The combined evidence supports a mechanism-oriented conclusion. Worker-specific joint routing/reordering adds sub-

stantial cache reuse on the high-locality workload, the gap increases at the heavier replicated load, and it contracts when locality is removed. This is stronger than a single-point winner but not a universal claim. The worker-blind ordering ablation separately shows that only cache-informed greedy ordering beats relevance order; merely imposing canonical or session-prefix order is harmful. Negative replications are equally informative: flat cache-aware variants are unresolved, overlap adaptation does not change outcomes even under drift, and beta’s old crossover was not robust to capacity calibration.

The principal threats to validity are:

- 1) **Protocol comparability.** Historical `per_worker_tree` TTFT excludes decision/reordering and cannot support an end-to-end speedup claim. Its prompt contains 1.9% more tokens because of a separator mismatch. The fixed harness must be rerun before comparing latency or exact cache fractions without this qualification.
- 2) **Two workers and a partial baseline.** At $N=2$, both hash candidates cover the whole pool. The local DualMap-inspired strategy also differs from the published system in mapping, migration, cost estimation, and load instrumentation. It must not be interpreted as official DualMap. Semantic pruning also cannot be evaluated when $k = N$.
- 3) **Synthetic, narrow workloads.** Every live trace is generated over the real TQuAD corpus, not captured production traffic. The locality axis has only two synthetic regimes, the main sweep one model and two GPUs, and the 4× point only one repetition. Generalisation requires an external trace such as ShareGPT or LMSYS.
- 4) **Metric and capacity history.** The original ordering and bookkeeping runs use 50,000 blocks; only ordering was replicated at 5,840. The ordering cache score is a per-request macro-average, while the system table aggregates tokens. Cross-table numerical comparisons must respect that distinction.
- 5) **Run reproducibility.** The repository lacks raw live JSONL outputs and a complete hardware/driver manifest. The new sweep scripts record configuration, arm order, lag, logs, and summary files, but the supplied numerical tables cannot yet be independently recomputed without rerunning both workers.
- 6) **Prototype calibration.** Semantic top- k and quality-protection parameters remain defaults. The CUSUM calibration uses one synthetic drift strength, and the per-worker adaptive path compares global adjacency against a within-session target; alarms do not control routing. Absence of separation is not evidence that every adaptive design is ineffective.
- 7) **Answer quality.** The new quality result validates the contains-gold metric under TQuAD only. The low-locality difference and `protect_top_k` need paired question-level tests; tree depth, reorder rate, and cache-prefix continuity are not yet instrumented.

VIII. ARTIFACT AVAILABILITY AND TEAM CONTRIBUTIONS

The complete source repository is available at <https://github.com/ubbx05/cache-aware-llm-router>. The submission archive contains the router, all routing strategies, replay and calibration scripts, `sweep_strategy.py`, `sweep_beta.py`, smoke tests, monitoring configuration, `SETUP.md`, dependency specification, and the $\LaTeX$ source and PDF of this paper; generated corpora, embeddings, traces, and live-run outputs are excluded. The raw TQuAD source is publicly available at <https://github.com/TQuad/turkish-nlp-qa-dataset>, and `bench/build_corpus.py` documents the deterministic text chunking and block-alignment transformation. Data fingerprints and trace manifests are supplied so an independently obtained copy can be checked against the experimental inputs.

Repository history reflects the following primary division of work. Umut Baran Boztaş developed the core proxy and replay pipeline, the per-worker/two-phase integration, and live deployment calibration. Saadet Cansu Baktıroğlu developed cache-measurement and adaptive, semantic, token-weighted, and quality-protection extensions and led the IEEE paper synthesis. Demir Doruk Dilek developed the replicated ordering and bookkeeping abstractions, CUSUM calibration, schedule checks, and reproducibility documentation. All authors reviewed the integrated system, experimental interpretation, and final artifact.

IX. CONCLUSION AND FUTURE WORK

At the calibrated speedup-eight operating point, worker-specific joint routing and ordering raises aggregate cached-token fraction from $66.4 \pm 0.4\%$ for flat cache-aware routing to $76.1 \pm 0.06\%$. Its advantage reaches 12.0 percentage points at the heaviest replicated load and contracts to 4.2 points under low locality. This is the paper’s main result: worker-aware histories provide measurable cache value, and the value follows the workload property the mechanism requires. The result does not yet establish end-to-end latency improvement because the historical two-phase timer and prompt separator differ from the one-phase arms.

Calibration and repetition also revise the surrounding story. Flat cache-aware variants and fixed/adaptive routing are unresolved; the semantic-labelled arm did not prune; the global greedy ordering ranking survives capacity calibration; the earlier beta crossover does not; and completion-time cache bookkeeping deteriorates with load. These findings argue for calibrated, repeated sweeps rather than selecting policies from one run.

Future work is therefore focused: (1) rerun the main and $4\times$ comparisons with matched serialization and the new end-to-end timer; (2) record reorder depth, changed-order rate, per-worker tree occupancy, and decision overhead to isolate why the joint policy wins; (3) repeat the locality result on an external workload and attach raw runs plus a complete environment fingerprint; and (4) test semantic pruning, the DualMap-inspired design, and quality protection on $N > 2$ workers with paired quality analysis.

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